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RESEARCH ARTICLE

A Hybrid Deep Learning Framework for Deepfake Detection Using Temporal and Spatial Features

FAZEEL ZAFAR¹, TALHA AHMED KHAN², SALAS AKBAR¹, MUHAMMAD TALHA UBAID³, SAMEENA JAVAID¹, AND KUSHSAIRY ABDUL KADIR⁴, (Senior Member, IEEE)

¹Department of Computer Sciences, School of Engineering and Applied Sciences, Bahria University, Karachi Campus, Karachi 75290, Pakistan

²Cybersecurity and Technological Convergence (CTC) Section, Malaysian Institute of Information Technology, Universiti Kuala Lumpur, Kuala Lumpur 53100, Malaysia

³Department of Information Technology, University of Central Punjab, Lahore 54000, Pakistan

⁴British Malaysian Institute, Universiti Kuala Lumpur, Kuala Lumpur 53100, Malaysia

Corresponding authors: Talha Ahmed Khan (talha@unikl.edu.my) and Kushsairy Abdul Kadir (kushsairy@unikl.edu.my)

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ABSTRACT The rise of deep-fake technology has sparked concerns as it blurs the distinction between fake media by harnessing Generative Adversarial Networks (GANs). This has raised issues surrounding privacy and security in the realm. This has led to a decrease in trust during online interactions; thus, emphasizing the importance of creating reliable methods for detection purposes. Our research introduces a model for detecting deepfakes by utilizing an Enhanced EfficientNet B0 structure in conjunction with Temporal Convolutional Neural Networks (TempCNNs). This approach aims to tackle the challenges presented by the evolving sophistication of deep-fake techniques. The system dissects video inputs into frames to extract features comprehensively by using Multi Test Convolutional Networks (MTCNN). This method ensures face detection and alignment by focusing on facial regions. To enhance the model's adaptability, to different scenarios and datasets we implement data augmentation techniques such as CutMix, MixUp and Random Erasing. These strategies help the model maintain its strength, against distortions found in deepfake content. The backbone of EfficientNet B0 utilizes Mobile Inverted Bottleneck Convolutions (MBConv) and Squeeze and Excitation (SE) blocks to enhance feature extraction by adjusting channels to highlight details effectively. A Feature Pyramid Network (FPN) facilitates the fusion of scale features capturing intricate details as well, as broader context. When tested on the FFIW 10 K dataset, which comprises 10,000 videos evenly split between manipulated content, the model attained a training accuracy of 91.5 % and a testing accuracy of 92.45 %, after 40 epochs. The findings showcase the model's proficiency, in identifying videos with precision and tackling the issue of class imbalances found in datasets – a valuable contribution, to advancing dependable deepfake detection solutions. Furthermore, the model achieves an impressive balance between accuracy and computational efficiency, attaining 92.45% testing accuracy with a lightweight computational cost of 0.45 GFLOPs, making it a highly practical choice for real-world deployment.

INDEX TERMS Deepfake detection, face detection and alignment, EfficientNet B0, temporal convolutional networks (TCNs), multi-task convolutional neural network (MTCNN), feature pyramid network (FPN), mobile inverted bottleneck convolutions (MBConv).

I. INTRODUCTION

As technology has significantly advanced in recent years, the challenges related to these advancements have also risen substantially in various fields. Among all other fields

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Deepfake is one of the major concern these days as deep-fake has put a question over the legitimacy of the media and information security. Deepfake uses namely Generative Adversarial Networks (GANs) [1], to generate hyper realistic synthesized/simulated media. Because of this artificial fabrication of digital content, establishing the difference between real and synthesized data has become significantly difficult.

These exaggeration in generation of fabricated media, both videos and images, have raised a major concern about media validity and its effect on societal norms, as deepfake generation may lead to procreation of false and misleading information as well as the loss of societal faith over digital content [2]. The advancements in GANs models and availability of larger datasets have helped in generation of remarkably realistic digital media that goes undetected by human eye, but as these models work on spatial and temporal fabrications, these fabrications can be detected using advanced techniques of deep learning models [3]. These concerning advancements in generation of false media demands for the establishment of advanced and effective deepfake detection algorithms. To overcome the challenges of deepfake detection, researchers has opted more advanced deep learning approaches that can detect spatiotemporal fabrications of media content more accurately [3]. Efficient Net belongs to the family of Convolutional Neural Networks (CNNs) that handles width, depth and resolution of the network to attain maximum accuracy. It is one of the many technologies being implemented for detection of deepfake media content, as Efficient Net uses far less computational cost then other techniques [4], making it one of the potential technique to detect fabricated media as it has capacity to extract precise information from images and videos. The advancement in production of concerning accurate fabricated media has presented us with a challenge far greater than just spatial domain, lack of temporal coherence in these fabricated images/videos is a very crucial aspect to be addressed. In recent years Temporal Neural Networks (TempCNNs) have become one of the main technique to handle sequential data. They offer significant advantage over Recurrent Neural Networks (RNNs) as TempCNNs maintains the relation between temporal scales in hierarchical manner [5]. When TempCNNs and Efficient Net are combined together they can provide a technique that can analyze spatiotemporal intricacies of deepfake media more accurately. In this paper, an integrated approach of Efficient Net and TempCNNs is proposed for deepfake detection, where Efficient Net is responsible for spatial feature extraction and TempCNNs for pattern recognition. By making most of the strength, proposed by both architectures this research aims for a better approach for detection real media from fake media. Given the increasing sophistication of deepfake generation techniques, there is a growing need for solutions that not only achieve high detection accuracy but also maintain computational efficiency. Traditional methods often sacrifice real-world applicability due to high computational costs. Our approach, however, presents a balanced trade-off, ensuring that high accuracy can be achieved without excessive resource consumption, making it a more viable solution for practical, real-world deployment.

II. LITERATURE REVIEW

The field of deepfake detection and related computer vision applications has seen significant advancements in recent years, with various models and techniques being developed

to improve accuracy and efficiency. The literature in this domain highlights the evolution of approaches, datasets, and performance metrics, demonstrating how research has progressively tackled the challenges posed by deep-fake technology.

Deepfake detection has leveraged Convolutional Neural Networks (CNNs) and hybrid models to achieve high detection accuracy. A 2024 study utilized Inception Net in a CNN-based comparative analysis, trained on a Kaggle dataset comprising 401 videos and 3,745 images post-augmentation, achieving an accuracy of 92.3% with moderate computational efficiency [6]. Similarly, a 2023 study applied Support Vector Machines (SVM) with Principal Component Analysis (PCA) on a custom dataset, transforming RGB images to YCbCr, leading to a 92.8% accuracy when using PCA, compared to 72.2% without PCA [8]. A 2025 study further enhanced face forgery detection by introducing attention-based feature learning and Optimized Long Short-Term Memory (OLSTM). Using datasets such as FaceForensics++, Deepfake Detection Challenge, Celebrity Deepfake, and Wild Deepfake, this approach achieved a cross-validated accuracy of 94.65%, albeit at a lower efficiency due to its reliance on OLSTM and ResNet-50 Multi-Head Attention (RN50MHA) [9].

Recent comparative analyses have shed light on the performance of various deepfake detection architectures. In 2024, researchers compared CNN+RNN, Deepfake Detection Network (DAFDN), Hybrid Inception ResNet v2, and Xception models on a comprehensive dataset. The study reported accuracy rates of 94.8% for CNN+RNN, 97.8% for DAFDN, and 98% for the Hybrid model, with efficiency varying based on computational demands [11]. Similarly, a 2022 study assessed multiple deepfake detection algorithms on FaceForensics++, reporting an accuracy of 96%, indicating that efficiency trade-offs are a key factor in model selection [20]. Further, Generative Adversarial Networks (GANs) have played a crucial role in image synthesis and deep-fake generation. A 2020 review of GAN variants in computer vision examined datasets such as LSUN-BEDROOM, CELEBA-HQ-128, and CIFAR-10, reporting a precision of over 90% for BigGAN, though computational costs were noted as a challenge [7]. Likewise, a 2018 study implemented Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN) using Residual Dense Blocks (RDB) without Batch Normalization (BN) layers. Despite achieving an RMSE score of 15.15% on the PIRM self-validation dataset, the model's high complexity made it less efficient [10].

On the other hand, Alternative deepfake detection methods have incorporated diverse techniques. A 2022 study employed MesoNet on FaceForensics++ and Celeb-DF, yielding an 88% accuracy while maintaining high efficiency due to its lightweight nature [13]. Another 2022 study focused on Point of Interest (POI) action detection using Multi-task Cascaded Convolutional Networks (MTCNN) and MesoNet on a customized dataset of Barack Obama, achieving an

impressive accuracy of 95.75% with high efficiency [14]. Additionally, a 2022 investigation combined MTCNN-based feature extraction with EfficientNet for real/fake classification, obtaining a lower accuracy of 62.5%, but benefiting from EfficientNet's lightweight nature [18].

A 2021 study compared different versions of EfficientNet on the Deepfake Detection Challenge (DFDC) dataset, achieving an accuracy of 82.9% while maintaining computational efficiency [19]. In contrast, a 2020 study implemented Discrete Cosine Transform (DCT) in an LSTM-based multimodal network, significantly reducing the feature space and improving efficiency despite achieving a relatively lower accuracy of 61.95% on DFDC [12]. Beyond deepfake detection, Temporal Convolutional Networks (TempCNNs) have been applied in remote sensing and epidemic prediction. A 2019 study on optical satellite imagery and reference data, such as Formosat 2 images and farmer declarations, achieved an accuracy of 93.45%, balancing performance with medium efficiency requirements [15]. Similarly, a 2021 study adapted TempCNN models for epidemic evolution prediction, using datasets with global mobility patterns, and reported training and validation RMSE scores of 4.1% and 2.7%, respectively [17].

The reviewed studies demonstrate the rapid advancements in deep-fake detection, computer vision applications, and related fields. While hybrid and attention-based models have achieved the highest accuracy, their computational complexity often limits efficiency. Lightweight models like MesoNet and EfficientNet balance accuracy with processing efficiency, making them suitable for real-time applications. Additionally, the use of TempCNNs and GAN-based models extends beyond deepfake detection, showcasing the versatility of deep learning in image processing and prediction tasks. Future research should focus on optimizing high-accuracy models for real-world deployment while minimizing computational overhead.

The Efficiency column highlights a tradeoff between efficiency and accuracy that how different techniques balance computational cost and detection accuracy in deepfake detection methods. While some methods achieve high accuracy, they often come with significant computational demands, making them less suitable for real-world, resource-constrained environments. For instance, models like GAN-based approaches and SRGANs prioritize accuracy, leading to high-performance results at the cost of requiring more computational resources. On the other hand, models like EfficientNet and MesoNet offer a better balance, providing competitive accuracy with relatively lower computational costs, making them more efficient for practical applications. This balance is crucial, especially as deep-fake detection needs to be fast and scalable without compromising performance. Models with a lighter computational cost like EfficientNet are particularly valuable for real-time applications, where efficiency is as critical as accuracy, and they enable better deployment in real-world settings with limited computational resources.

TABLE 1. Literature review.

Year	Technique	Dataset	Metrics	Efficiency
2024 [6]	Deepfake detection using Inception Net with CNN-based comparative analysis	Kaggle (401 videos, 3745 images after augmentation)	Accuracy: 92.3%	Medium (CNN-based model balances accuracy with computational efficiency)
2020 [7]	Review of GAN variants in CV	LSUN-BEDROOM M CELEBA-HQ-128 CIFAR 10	Precision BIGGAN: >90%	Medium (GANs are computationally expensive)
2023 [8]	Fake face detection using SVM and PCA with edge extraction	Custom dataset (Preprocessed RGB to YCbCr images)	Accuracy: 92.8% (SVM with PCA), 72.2% (SVM alone)	Medium (SVM with PCA enhances accuracy while maintaining efficiency)
2025 [9]	Face forgery detection using attention-based feature	FaceForensics++, Deepfake Detection Challenge, Celebrity Deepfake, Wild Deepfake	Cross Validated Accuracy: 94.65%	Low (OLSTM and RN50MHA provide high detection rates with attention-based learning)
2018 [10]	Implementing SRGAN using RDBB without BN Layers SRResNet	Training Data : DIV2K, Flickr2K, OutdoorSceneTraining(OST) Testing Data: PIRM self-Val Dataset	RMSE score: 15.15%	Low (High complexity due to residual dense blocks)
2024 [11]	Comparative analysis of deepfake detection models (CNN+RNN, DAFDN, Hybrid Inception ResNet)	Comprehensive dataset	CNN+RNN: 94.8%, DAFDN: 97.8%, Hybrid: 98%	Low (Efficient compared to models like Xception but still resource-intensive)

TABLE 1. (Continued.) Literature review.

2020 [12]	Discrete Cosine transform – to create multimodal network. (LSTM)	Facebook Deepfake Detection Challenge (DFDC) ---Public	Accuracy: 61.95%	High (DCT reduces feature space, making it efficient)
2022 [13]	Deepfake Detection using MesoNet	FaceForensics++ Celeb-DF	Accuracy: 88%	High (MesoNet is lightweight and optimized for detection)
2022 [14]	POI's action detection using MTCNN and MesoNet	Customized Dataset of Barack Obama	Accuracy: 95.75%	High (Combination of MTCNN and MesoNet ensures efficiency)
2019 [15]	Remote Sensing of SITS via TempCNNs	Optical Satellite Data :Formsat 2 Images Reference Data: Farmer's Declaration 2006, Field Campaigns 2006,	Accuracy: 93.45%	Medium (Temporal CNNs require significant training data)
2022 [16]	Deepfake detection via CNNs on Haar Cascade processed images	DFDC (60 Clips)	Accuracy: 98%	Low (Haar Cascades require pre-processing and increase complexity)
2021 [17]	TempCNNs adapted model for epidemic evolution prediction	World Wide Dataset, French Dataset, World Wide Dataset with mobility, French Dataset with mobility	Whole Dataset Training RMSE: 4.1% Whole Dataset Validation	Medium (Temporal models need large datasets)
2022 [18]	MTCNN base feature extractions and Efficient Net for real/fake classification	FaceForensics++	Accuracy: 62.5%	High (EfficientNet is lightweight and optimized)
2021 [19]	Comparative study between versions of EfficientNet	DFDC	Accuracy: 82.9%	High (EfficientNet provides accuracy with lower complexity)

TABLE 1. (Continued.) Literature review.

2022 [20]	Comparative study between Deepfake detection algorithms	FaceForensics++	Accuracy: 96%	Medium (Accuracy varies based on computational trade-offs)
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III. METHODOLOGY

The video input given to the system first split into discrete segments before being processed. The step of Frame extractions and preprocessing is performed in order to get more features extracted by examining and processing individual segments of the videos independently. Video segmentation is done by splitting the frames into discrete segments. Video V is segmented in N frames each segment represented by S_i :

$$V = \{S_1, S_2, S_3 \dots S_N\} \quad (1)$$

Each segment is preprocessed, which include normalization $N(S_i)$, resizing $R(S_i)$, and augmentation $A(S_i)$

$$S'_i = A(R(N(S_i))) \quad (2)$$

In order to detect face, its alignment and landmarks of the face, Multi-Test Convolutional Neural Network is used. Multi-Test CNN works by mapping any image X to a set of facial landmarks L , $L = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ where n is the number of detected land marks, with base CNN functions, to make sure filtering off the background noise and concentrates the face only to analyze the pertinent facial regions.

$$L = F_{\text{CNN}}(X) \quad (3)$$

The alignment process makes sure consistency between each segment of the video. The methods such as data augmentation are used to boost the model's generalization and variance. By incorporating artificial distortion methods such as CutMix, MixUp, for any given data, along with RandomErasing strengthen model's resistance to different kind of modifications and distortions often encountered in deepfakes. From the segments of the input video, the framework can now work better for varying positions and occlusions, all thanks to the performed augmentations. The feature extraction method of EfficientNet B0 processes the segments of the video once reorganization and preprocessing is done. The model also has adaptive input feature that modifies the output of the input segments according to their output complexity. Since more intricate segments maintain greater resolution for improved precision, easier segments have to be downscaled to minimize computing expenses.

The Stem Block, which applies a 3×3 convolution with 32 filters, along with batch normalization (BN),

$$Z = \text{BN}(\text{Conv}(W, X)) \cdot \text{Swish}(Z) \quad (4)$$

as well as Swish activation for effective irregularities are the next steps in the machine's processing of the segments.

$$\text{Swish}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \quad (5)$$

With less resolution yet a greater representation of the feature, the feature maps produced by this layer are more detailed. The Backbone method EfficientNet B0 that is made up of Mobile Inverted Bottleneck Convolutions (MBConv) processes the individual segments after that. Individual channel of the input data X goes through a filter by MBConv block and then join the outputs X' utilizing a point-by-point 1×1 convolution in order to reduce the workload of the computation through depth wise distinct convolution.

$$X' = X \cdot \sigma(W2 \cdot \text{ReLU}(W1 \cdot \text{GAP}(X))) \quad (6)$$

where $W1$ and $W2$ are weight matrices, and σ is the sigmoid function.

Every single MBConv block contains a squeeze-and-excitation (SE) block that recalibrates the characteristics of the channels and enables the framework to concentrate on significant characteristics. This makes sure that unnecessary characteristics are hidden and the important details which set both real and fake apart are emphasized. Initially the SE block works by pressing down the spatial dimensions of the characteristic maps through global pooling and then employs an activation function of sigmoid along with fully connected layers in order to excite the feature channel. To cater the problem of the vanishing gradient and encourage secure training, residual links are included into the MBConv blocks' deeper layers. Better understanding of deeper depictions is made possible by such connections. The addition of Multi-Scale Feature Fusion through the Feature Pyramid Network (FPN) is a significant structural improvement. The FPN combines data through initial, middle, and deeper layers of the backbone EfficientNet B0 to bring together features from different layers. The contextual data that is more broadly relevant, like irregularities in facial shape, is caught as well as minute details like subtle motions of the face due to this broad scale fusion. The framework's capacity to identify minute deepfake irregularities, including lip-sync errors or minute changes in emotions of the face, is enhanced by this combination. After the obtaining of features is finished, a TempCNN is trained on the obtained segment feature stream.

Because deepfake modifications frequently result in inconsistent motion and facial expressions across time, the TCN is in charge of simulating the temporal relationships between segments. The learning network may develop dependencies that are long-range yet preserve secure flow of gradients during training thanks to these Residual TCN Layers. The significant temporal data is kept in the framework by incorporating the residing links. An additional essential component of TCN is adaptive dilation. Adaptive dilation rapidly alters the TCN's responsive field to accommodate changes that occur in temporal inconsistency's scale of time. This is necessary to record both transient and persistent discrepancies in

the movements of the face present in the video.

$$H_t = \sum_{i=0}^{k-1} W_i \cdot F_{t-id} \quad (7)$$

where W_i are learnable weights, k is the filter size, F is a feature vector and d is the dilation rate. TCN incorporates Gated Linear Units (GLUs) to regulate the data flow across the system.

$$O = X \cdot \sigma(G) \quad (8)$$

GLUs improve the framework's capacity to identify minute spatial oddities, like aberrant actions between segments or variations in movement of the face that could be signs of modification. After being processed by a layer of linear classification, the output of TCN yields just one logit which reflects after characterization of the input video either real or fake. The temporal discrepancies and spatial characteristics that were taken out of the segments are the basis for the ultimate categorization choice. Calculated binary cross-entropy loss function training assists the framework concentrated on complex instances in which deepfake modifications tend to be subtle and harder to spot. By preventing the framework from over fitting to instances which are simple to classify, the calculating system raises the general precision of deepfake detection. The last stage involves applying subsequent processing methods to the series of forecasts created throughout the video segments, including confidence achieving and voting by majority. Employing the vote of the majority, the forecasts to each of the segments are combined to determine the last categorization for the whole footage. To show how confident the algorithm is in identifying it, a score for confidence is also produced. More thorough choices are facilitated by this, particularly in ambiguous situations in which categorization is unclear. The last real and fake categorization is made along with the corresponding confidence score are output by the system.

A. ENHANCED EFFICIENTNET B0 ARCHITECTURE AND FEATURE EXTRACTION

A key element of the suggested method is the Enhanced EfficientNet B0 design, the basic structure is the mobile inverted bottleneck convolution (MBConv) module, which also introduces the compression and excitation network's attention concept (Squeeze-and-Excitation Network, SENet) [4], which is in charge of obtaining fine-grained spatial features from segments of the video. Starting with an Input Layer, variable data scaling is incorporated. In accordance with the amount of detail of the supplied segments, this scaling process continually alters the pixel size of the segments. Although intricate segments are kept at greater resolutions for deeper examination, segments with easier information are downsampled to minimize computation overhead costs. The optimization of the segments for precision and computational effectiveness is ensured by this preliminary processing phase. Following that, 32-filter with a convolution of 3×3 is applied by the Stem Block, which then follows Batch Normalization (BN) and Swish stimulation to produce map features

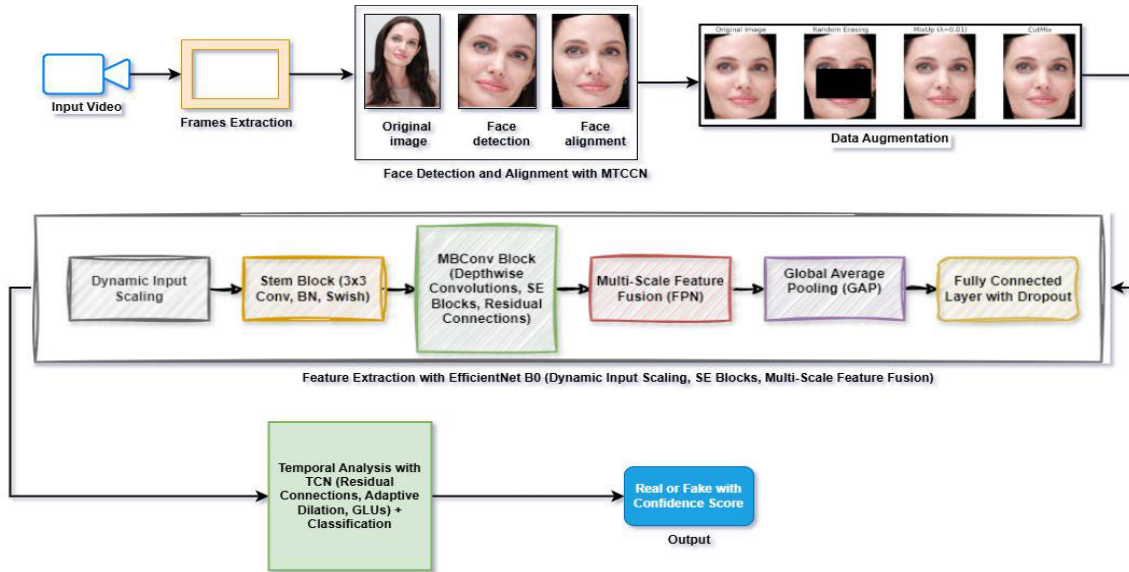


FIGURE 1. The main architecture.

with improved visualization of features and lower resolution. The Backbone architecture of EfficientNet B0, that includes Mobile Inverted Bottleneck Convolutions (MBConv), that serves as the foundation of EfficientNet [21], processes such data. To minimize the amount of computation required, the MBConv eliminates use depth-wise distinct convolutions. A 1×1 point-by-point convolution is then used for merging the resulting channels. To reset the features of the channels, squeeze-and-excitation (SE) section have been included to every MBConv block. The framework can eliminate undesirable features and highlight more significant ones thanks to this adjustment procedure. For SE blocks to function, feature maps' spatial dimensions are first compressed via global average pooling, after which fully connected layers along with activation sigmoid function is used as a stimulation stage. The capacity of the framework to distinguish minute variations among genuine and fake segments is enhanced by this priority. To guarantee secure training of complex visualizations and alleviate the gradient disappearing issue, remaining connections are added to the backbone's deeper layers. Deeper learning is made possible through such residual interactions, which also protect significant data from previous layers. The Feature Pyramid Network (FPN) facilitates Multi-Scale Feature Fusion, which improves the framework even more. The EfficientNet B0 backbone's multiple layers' characteristics are combined by the FPN to ensure that certain details and more comprehensive context-related data are recorded. The simulator can identify minute adjustments in deep-fake footage, such as irregularities in facial contours and slight movements of the face, thanks to that multi-scale combination of features. The algorithm can cope with high quantities deep-fake detection tasks because Dynamic Input Scaling makes certain that easier segments are processed faster while segments with more

information are handled at greater resolution as shown in figure 2.

In the above methodology, spatial and temporal features are combined through a multi-step process to enhance deepfake detection. Spatial features are extracted from each video segment using EfficientNet B0, which captures detailed facial landmarks and expressions, while the Feature Pyramid Network (FPN) integrates features from different layers for both fine and broader contextual details. Temporal features are then analyzed using a Temporal Convolutional Network (TCN), which processes the sequence of spatial features to detect inconsistencies in facial motion or expression over time. By combining these spatial and temporal insights, the system can detect deep-fake artifacts that may not be apparent in individual frames, ensuring more accurate classification by considering both static facial details and dynamic changes across the video.

B. GLOBAL AVERAGE POOLING, CLASSIFICATION, AND TEMPORAL ANALYSIS

Following the extraction of pertinent spatial features to the segments by the backbone, a Global Average Pooling (GAP) Layer processes the resultant data. Individual map of features is reduced to just one scalar value by the GAP layer, which also decreases the spatial dimensions of the mapped features. This procedure guarantees every segment's spatial data is reduced in a summary vector. After that, an entirely connected layer via regularization for dropouts receives the output from the GAP layer. By independently eliminating neurons throughout training, dropout guarantees that the algorithm applies well to new data and prevents overfitting. A temporal convolutional network (TCN) receives the highly dense feature vector produced by the fully connected layer to perform temporal evaluation. To represent long-range time-dependent

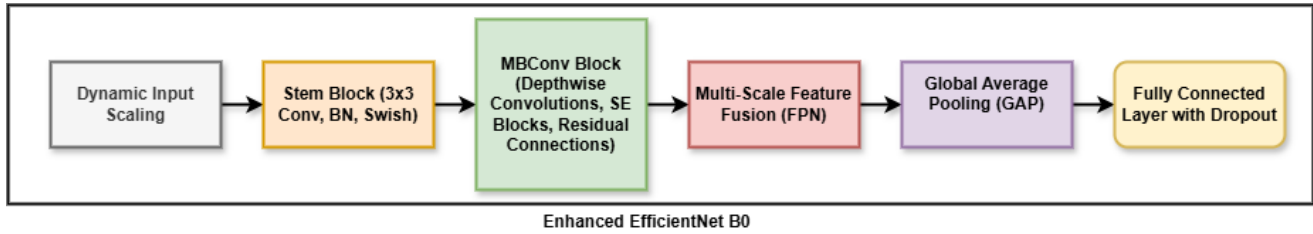


FIGURE 2. Enhanced efficientnet B0.

relationships among segments, the Temporal Convolutional Network (TCN) procedures the feature vector order derived from segments [22]. While training, the system assists in learning such temporal connections by the Residual TCN Layers, which also preserve secure gradient flow. Furthermore, the TCN's responsive field is constantly modified via Adaptive Dilation, allowing it to detect short- and long-term together temporal discrepancies in the clip. Lastly, TCN uses Gated Linear Units (GLUs) to control the transfer of data, enabling the network to concentrate on minute temporal deviations like erratic movements of the face or inconsistent motion. The algorithm can reliably and effectively identify deep-fake modifications on a variety of video data thanks to the integration of improved obtaining of characteristics, multi-scale fusion, and temporal evaluation.

IV. MATERIALS AND METHODS

A. DATASET

The dataset we've used for our experiment is FFIW 10K [23] which consists of 10,000 videos, with an equal distribution of 5,000 fake and 5,000 real videos. This ensures that there is no class imbalance and this also sets our dataset apart from many other deepfake datasets, such as Deepfake TIMIT [24], Celeb-DF [25], FaceForensics++ [2], and DFDC [26]. For example, the Deepfake TIMIT dataset contains about 320 real and 640 fake, Celeb-DF contains 590 real and 5639 fake videos, FaceForensics++ contains 1000 real and 4000 fake videos, and DFDC contains 104500 fake videos and 23654 real videos. These class imbalances can introduce bias during model training. In comparison to these datasets, FFIW 10K provides a more balanced dataset which ensures the model makes it more reliable and generalized, and a good benchmark for training deepfake detection models. This dataset also contains high-quality videos which enhances its utility for research in computer vision and deep learning.

The dataset used in this study, FFIW 10K, consists of 10,000 videos, with an equal distribution of 5,000 real and 5,000 fake videos, ensuring no class imbalance. This balanced dataset sets it apart from other datasets like Deepfake TIMIT, Celeb-DF, FaceForensics++, and DFDC, which typically suffer from unequal distributions, often skewed toward fake videos. For instance, Deepfake TIMIT has more fake videos, which could introduce bias during training. The balanced nature of FFIW 10K helps the model train

on a more reliable dataset, improving its generalization and reducing bias. To further enhance adaptability and robustness, several data augmentation techniques were applied, including CutMix, MixUp, and Random Erasing. CutMix combines two different images by cutting and pasting them together, which helps the model learn robust features by exposing it to diverse data combinations. MixUp generates synthetic samples by averaging two images, encouraging the model to learn smoother decision boundaries and reducing overfitting. Random Erasing randomly selects and erases parts of an image, making the model focus on other important regions, which helps it recognize features even when parts of the input are missing or altered. These augmentation techniques, combined with the balanced nature of FFIW 10K, ensure that the model is more adaptable to varying deepfake manipulations and can generalize better to real-world scenarios where videos may have subtle changes or distortions.

B. IMPLEMENTATION DETAILS

The system that was used for training and testing the model had an AMD Ryzen 5 5600 along with an NVIDIA GeForce GTX 1060 6GB and 32GB RAM. Model implementation was done on Windows 10 OS with Python 3.9.13 and PyTorch. For activation function, Rectified Linear Unit (ReLU) was used in the hidden layers of our TempCNN, and for the final output we applied a Sigmoid function to convert the logits in probabilities which were then thresholded at 0.5 to generate binary predictions.

The learning rate was set to 0.000001 initially during training, with a batch size of 16 with Adam optimization function and "Random Normal Initialization" method for weights initialization, which is stated in this equation:

$$(W \sim N((\mu, \sigma^2))) \quad (9)$$

where W is the weight matrix being initialized, and $N((\mu, \sigma^2))$ denotes a normal distribution with mean μ and variance σ^2 . The loss function was Binary Cross-Entropy with Logits Loss, which stated in this equation:

$$\text{BCEWithLogitsLoss}(y, \hat{y}) = \max(\hat{y}, 0) - \hat{y} \cdot y + \log(1 + \exp(-|\hat{y}|)) \quad (10)$$

where $\max(\hat{y}, 0)$ ensures numerical stability, $\hat{y}$ is the predicted output. The term $\log(1 + \exp(-|\hat{y}|))$ is derived from the sigmoid function. This ensures that the logits $\hat{y}$ are converted

into a probability-like value without explicitly calculating the sigmoid value.

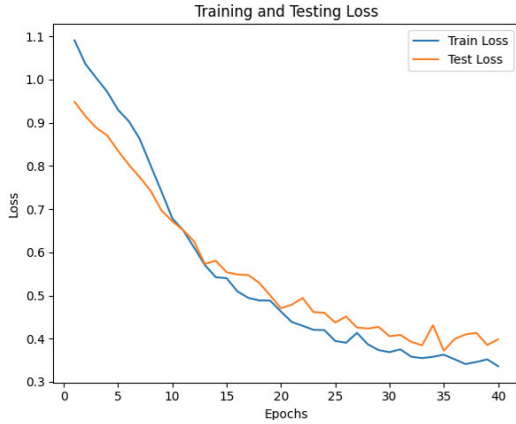


FIGURE 3. Loss graph of training and testing.



FIGURE 4. Training and validation accuracy graph.

The proposed model achieved 91.5% training accuracy and 92.45% testing accuracy at 40 epochs as shown in Figure 4. Our model is able to detect real and fake videos with high accuracy.

The confusion matrix presented in Figure 5 summarizes the performance of our model in detecting real and fake videos during validation. The key elements of the confusion matrix are explained below:

True Positives (TP): 932 videos classified correctly as REAL

True Negatives (TN): 905 videos classified correctly as FAKE

False Positives (FP): 39 REAL videos classified incorrectly as FAKE

False Negatives (FN): 111 FAKE videos classified incorrectly as REAL

The accuracy of the model can be calculated using the following formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (11)$$

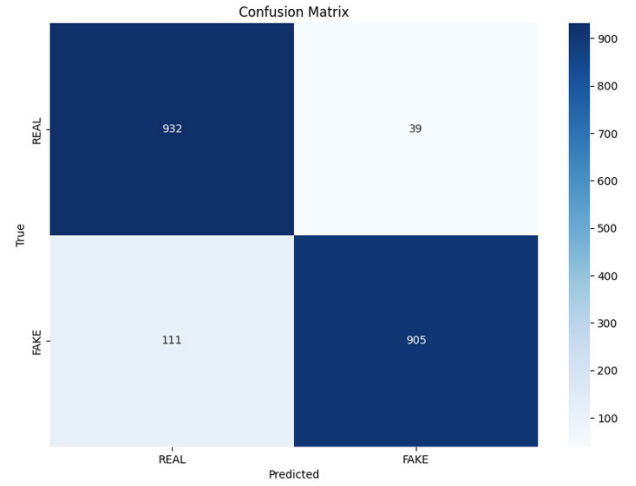


FIGURE 5. Confusion matrix of our model.

where: TP is number of True Positives = 932

TN is number of True Negatives = 905

FP is number of False Positives = 39

FN is number of False Negatives = 111

Using these values on the formula:

$$\begin{aligned} \text{Accuracy} &= \frac{932 + 905}{932 + 905 + 39 + 111} \\ &= 0.924509 \times 100 = 92.45\% \end{aligned}$$

C. COMPUTATIONAL EFFICIENCY

One of the standout features of our proposed deepfake detection model is its lightweight computational footprint, achieving an efficiency of 0.45 GFLOPs (Giga Floating Point Operations per Second). This efficiency is the result of several key architectural optimizations:

1) EFFICIENTNET B0 BACKBONE

Our model utilizes the EfficientNet B0 architecture, which employs Mobile Inverted Bottleneck Convolutions (MBConv) to significantly reduce the number of operations without compromising feature extraction quality. The depth wise separable convolutions within MBConv blocks further minimize computational load by reducing redundant operations [27].

2) DYNAMIC INPUT SCALING

The model adapts input resolution based on the segment's complexity—higher-resolution processing for intricate segments and lower-resolution processing for simpler ones. This ensures computational resources are allocated efficiently, preventing unnecessary overhead.

3) FEATURE PYRAMID NETWORK (FPN) FOR MULTI-SCALE FUSION

By incorporating an FPN, our framework merges spatial information across different layers, reducing the need for

excessive computation while still capturing crucial deepfake artifacts. This method optimizes feature representation without requiring extra convolutional layers, keeping operations minimal.

4) TEMPORAL CONVOLUTIONAL NETWORK (TCN) WITH ADAPTIVE DILATION

The TCN efficiently models temporal inconsistencies by adjusting its receptive field dynamically. Instead of brute-force computation over long sequences, adaptive dilation selectively expands the network's focus to capture essential temporal dependencies while skipping redundant calculations.

5) GATED LINEAR UNITS (GLUS) FOR EFFICIENT PROCESSING

GLUs regulate the flow of information, ensuring that only the most relevant features are processed while filtering out redundant activations. This selective computation contributes to reducing the overall GFLOPs count.

6) WHY 0.45 GFLOPS MATTERS

By achieving a computational cost of 0.45 GFLOPs, our model strikes a perfect balance between performance and efficiency. This makes it suitable for real-world applications, including real-time deepfake detection on consumer-grade hardware and mobile devices, where processing power is often limited. Unlike many deep learning models that require high-end GPUs for inference, our approach ensures that deepfake detection remains accessible, fast, and scalable.

D. RESULTS AND EXPERIMENTS

In this section we're comparing the performance of our proposed model with two other, similar models that were trained as an experiment with different parameters. The models were evaluated based on their precision, recall, f1-score and accuracy values, which are stated in Eq (12), Eq (13) and Eq (14). The learning rate of 0.000001 (1e-6) was held constant across all the models.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$F1 - \text{Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

The first model which was trained with batch size of 12 for 40 epochs had the lowest performance, achieving an accuracy of 88.57%. The recall value for the REAL class was much higher than the FAKE class which indicated that the model was biased towards the REAL class, which affects model reliability.

The second model was trained with batch size of 10 for 30 epochs. This improved the recall value for FAKE class while also achieving higher accuracy of 90.29%. This model made more balanced predictions than the previous model.

TABLE 2. Classification report of 1st model.

	Precision	Recall	F1-Score
REAL	0.820339	0.984741	0.984741
FAKE	0.981413	0.788845	0.874655
Accuracy	0.885757	0.885757	0.885757
Macro Average	0.900876	0.886793	0.884854
Weighted Average	0.901727	0.885757	0.884746

TABLE 3. Classification report of 2nd model.

	Precision	Recall	F1-Score
REAL	0.878365	0.92542	0.901279
FAKE	0.927846	0.882126	0.904408
Accuracy	0.902869	0.902869	0.902869
Macro Average	0.903105	0.903773	0.902843
Weighted Average	0.904139	0.902869	0.902909

Finally, the third model performed the best out of the previous two models, achieving a higher accuracy of 92.45% and the precision and recall values for both REAL and FAKE classes were almost similar, which demonstrated a better balance.

TABLE 4. Classification report of 3rd model.

	Precision	Recall	F1-Score
REAL	0.893576	0.959835	0.925521
FAKE	0.958686	0.890748	0.923469
Accuracy	0.924509	0.924509	0.924509
Macro Average	0.926131	0.925292	0.924495
Weighted Average	0.926869	0.924509	0.924472

V. CONCLUSION

The proposed study develops a deepfake video detection approach through an Enhanced EfficientNet B0 architecture combined with Temporal Convolutional Networks that handle the challenges of deepfake technologies. Our work involved segmenting the input videos to extract features, while data augmentation methods were used to improve both performance and adaptability for the models. During training, it achieved 91.5% accuracy and later 92.45% during the test runs on the FFIW 10k dataset, which clearly demonstrates the efficiency of this network in detecting manipulated videos.

A key contribution of this paper is the development of a computationally efficient model that achieves high detection

accuracy while maintaining a lightweight computational cost of just 0.45 GFLOPs. This balance between performance and efficiency ensures that the model is not only accurate but also practical for real-world applications, including mobile and low-power edge devices. Our findings emphasize that balanced datasets and advanced architecture play a crucial role in developing reliable content detection methods within the fast-growing domain of deepfake detection.

The improvement in deepfake detection could be achieved by increasing the dataset with different deepfake methods and real-world scenarios to make the model adaptable for the future. The inclusion of data sources such as audio and text may result in a full-scale detection system that gathers information from different sources. Additionally, applying adversarial training methods could further strengthen the model against emerging deep-fake generation techniques. Furthermore, as noted earlier, enhancing both efficiency and precision will be crucial in ensuring the model's real-time performance. Researchers and decision-makers must work together constantly to ensure that AI is effectively applied to mitigate the risks associated with deepfake technology, while also addressing ethical concerns and promoting best practices.

REFERENCES

- [1] F. B. Aissa, M. Hamdi, M. Zaid, and M. Mejdoub, "An overview of GAN-DeepFakes detection: Proposal, improvement, and evaluation," *Multimedia Tools Appl.*, vol. 83, no. 11, pp. 32343–32365, Sep. 2023, doi: [10.1007/s11042-023-16761-4](https://doi.org/10.1007/s11042-023-16761-4).
- [2] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," 2019, *arXiv:1901.08971*.
- [3] O. de Lima, S. Franklin, S. Basu, B. Karwoski, and A. George, "Deepfake detection using spatiotemporal convolutional networks," 2020, *arXiv:2006.14749*.
- [4] V.-T. Hoang and K.-H. Jo, "Practical analysis on architecture of EfficientNet," in *Proc. 14th Int. Conf. Hum. Syst. Interact. (HSI)*, Jul. 2021, pp. 1–4, doi: [10.1109/HSI52170.2021.9538782](https://doi.org/10.1109/HSI52170.2021.9538782).
- [5] C. Lea, R. Vidal, A. Reiter, and G. D. Hager, "Temporal convolutional networks: A unified approach to action segmentation," in *Proc. Comput. Vis.-ECCV Workshops*, G. Hua and H. Jégou, Eds., Cham, Switzerland: Springer, Jan. 2016, pp. 47–54, doi: [10.1007/978-3-319-49409-8_7](https://doi.org/10.1007/978-3-319-49409-8_7).
- [6] D. V. Nagagopiraju, K. Ayyappa, P. Anshulalitha, J. Srikanth, and K. T. Teja, "A efficient deep fake face detection using deep inception net learning algorithm," *Turkish J. Comput. Math. Educ.*, vol. 15, no. 1, pp. 138–141, Mar. 2024.
- [7] Z. Wang, Q. She, and T. E. Ward, "Generative adversarial networks in computer vision: A survey and taxonomy," 2019, *arXiv:1906.01529*.
- [8] H. Sabah Abdul kareem and M. S. Mahdi Altaei, "Detection of deep fake in face images based machine learning," *Al-Salam J. Eng. Technol.*, vol. 2, no. 2, pp. 1–12, Feb. 2023.
- [9] R. R. Sekar, T. D. Rajkumar, and K. R. Anne, "Deep fake detection using an optimal deep learning model with multi head attention-based feature extraction scheme," *Vis. Comput.*, vol. 41, no. 4, pp. 2783–2800, Mar. 2025.
- [10] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, C. Change Loy, Y. Qiao, and X. Tang, "ESRGAN: Enhanced super-resolution generative adversarial networks," 2018, *arXiv:1809.00219*.
- [11] A. Rajeev and P. Raviraj, "Performance evaluation of deep learning models for detecting deep fakes," *Int. J. Systematic Innov.*, vol. 8, no. 1, pp. 49–62, 2024.
- [12] J. K. Lewis, I. E. Toubal, H. Chen, V. Sandesera, M. Lomnitz, Z. Hampel-Arias, C. Prasad, and K. Palaniappan, "Deepfake video detection based on spatial, spectral, and temporal inconsistencies using multi-modal deep learning," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, Oct. 2020, pp. 1–9, doi: [10.1109/AIPR50011.2020.9425167](https://doi.org/10.1109/AIPR50011.2020.9425167).
- [13] Z. Xia, T. Qiao, M. Xu, X. Wu, L. Han, and Y. Chen, "Deepfake video detection based on MesoNet with preprocessing module," *Symmetry*, vol. 14, no. 5, p. 939, May 2022, doi: [10.3390/sym14050939](https://doi.org/10.3390/sym14050939).
- [14] Q. Jaleel and I. Hadi, "Facial action unit-based deepfake video detection using deep learning," in *Proc. 4th Int. Conf. Current Res. Eng. Sci. Appl. (ICCRESA)*, Dec. 2022, pp. 228–233, doi: [10.1109/iccreesa57091.2022.10352085](https://doi.org/10.1109/iccreesa57091.2022.10352085).
- [15] C. Pelletier, G. Webb, and F. Petitjean, "Temporal convolutional neural network for the classification of satellite image time series," *Remote Sens.*, vol. 11, no. 5, p. 523, Mar. 2019, doi: [10.3390/rs11050523](https://doi.org/10.3390/rs11050523).
- [16] S. R. Adnan and H. A. Abdulbaqi, "Deepfake video detection based on convolutional neural networks," in *Proc. Int. Conf. Data Sci. Intell. Comput. (ICDSIC)*, Nov. 2022, pp. 65–69, doi: [10.1109/ICDSIC56987.2022.10075830](https://doi.org/10.1109/ICDSIC56987.2022.10075830).
- [17] L. Mohimont, A. Chemchem, F. Alin, M. Krajecki, and L. A. Steffene, "Convolutional neural networks and temporal CNNs for COVID-19 forecasting in France," *Appl. Intell.*, vol. 51, no. 12, pp. 8784–8809, Dec. 2021, doi: [10.1007/s10489-021-02359-6](https://doi.org/10.1007/s10489-021-02359-6).
- [18] T.-A. To, H.-C. Luong, N.-T. Nguyen, T.-T. Nguyen, M.-T. Tran, and T.-L. Do, "Deepfake detection using EfficientNet: Working towards dense sampling and frames selection," in *Proc. RIVF Int. Conf. Comput. Commun. Technol. (RIVF)*, Dec. 2022, pp. 612–617, doi: [10.1109/RIVF55975.2022.10013900](https://doi.org/10.1109/RIVF55975.2022.10013900).
- [19] A. A. Pokroy and A. D. Egorov, "EfficientNets for DeepFake detection: Comparison of pretrained models," in *Proc. IEEE Conf. Russian Young Researchers Electr. Electron. Eng. (ElConRus)*, Jan. 2021, pp. 598–600, doi: [10.1109/ElConRus51938.2021.9396092](https://doi.org/10.1109/ElConRus51938.2021.9396092).
- [20] L. Deng, H. Suo, and D. Li, "Deepfake video detection based on EfficientNet-V2 network," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–13, Apr. 2022, doi: [10.1155/2022/3441549](https://doi.org/10.1155/2022/3441549).
- [21] M. Shaikh, Q. A. Arain, I. F. Siddiqui, and H. A. Shaikh, "Automated classification of pneumonia from chest X-ray images using deep transfer learning EfficientNet-B0 model," in *Proc. 14th Int. Conf. Math., Actuarial Sci., Comput. Sci. Statist. (MACS)*, Nov. 2022, pp. 1–6, doi: [10.1109/MACS56771.2022.10022700](https://doi.org/10.1109/MACS56771.2022.10022700).
- [22] B. A. Kumar and N. K. Misra, "Masked face age and gender identification using CAFFE-modified MobileNetV2 on photo and real-time video images by transfer learning and deep learning techniques," *Expert Syst. Appl.*, vol. 246, Jul. 2024, Art. no. 123179.
- [23] T. Zhou, W. Wang, Z. Liang, and J. Shen, "Face forensics in the wild," 2021, *arXiv:2103.16076*.
- [24] Idiap Res. Inst., Artif. Intell. for Soc. *DeepfakeTIMIT*. Accessed: Nov. 11, 2024. [Online]. Available: https://www.idiap.ch/en/scientific-research/data/deepfaketimit/index_html
- [25] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for DeepFake forensics," 2019, *arXiv:1909.12962*.
- [26] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. Canton Ferrer, "The DeepFake detection challenge (DFDC) dataset," 2020, *arXiv:2006.07397*.
- [27] S. Javid and S. Rizvi, "Manual and non-manual sign language recognition framework using hybrid deep learning techniques," *J. Intell. Fuzzy Syst.*, vol. 45, no. 3, pp. 3823–3833, Aug. 2023.
- [28] I. Ali Hasani and O. Arif, "Pose calibrated feature aggregation for video face set recognition in unconstrained environments," *IEEE Access*, vol. 12, pp. 156337–156346, 2024, doi: [10.1109/ACCESS.2024.3481636](https://doi.org/10.1109/ACCESS.2024.3481636).



FAZEEL ZAFAR received the B.S. degree in computer science with specialization in computer vision and deep learning from Bahria University, Karachi Campus, Pakistan. Currently, he is a Freelance Machine Learning Engineer, collaborating with various multi-national clients. His research interests include computer vision, machine learning, deep learning, and generative AI.



TALHA AHMED KHAN received the B.Eng. degree in electronics and the M.Eng. degree in telecommunications from UIT, the P.G.D. degree in cyber security, the Postdoctorate degree in machine learning from MMU, Cyber Jaya, Malaysia, and the Ph.D. degree in AI from UniKL, Kuala Lumpur, Malaysia. He is currently associated with the Cybersecurity and Technological Convergence Section, MIIT, UniKL. Previously, he was associated with the Department of Artificial Intelligence, Bahria University, Karachi Campus, as the Head of the Department. He has published numerous high scientific impact factor research papers in well-reputed journals, IEEE conferences, Scopus indexed journals, and HEC recognized journals.

He has presented her research at numerous prestigious conferences worldwide. Her research interests include computer vision, machine learning, deep learning, and ubiquitous computing.



SAMEENA JAVAID received the Ph.D. degree in computer science from Bahria University, Karachi Campus, Pakistan. She has previously served Computer Science Department, Bahria University Karachi Campus. She is currently doing her postdoctoral research with Multimedia University, Malaysia, further enhancing her expertise in advanced machine learning techniques and real-world applications of computer vision. She is the author of several impact factor publications and

has presented her research at numerous prestigious conferences worldwide. Her research interests include computer vision, machine learning, deep learning, and ubiquitous computing.

SALAS AKBAR received the master's degree in computer science and information technology (MCIT) from the Ned University of Engineering and Technology, in 2022. Currently, she is a Senior Lecturer with the Program of Bachelor's in Artificial intelligence, Department of Computer Sciences, School of Engineering and Applied Sciences. She is also looking for opportunities to share her research ideas and deepen her knowledge. She hopes to engage with other researchers, gain valuable feedback, and contribute to meaningful discussions in her field. Her research interests include machine learning, deep learning, and natural language processing.



MUHAMMAD TALHA UBAID received the M.S. degree in computer science with specialization in computer vision domain from the University of Central Punjab (UCP), Lahore, Pakistan. He was associated with the University of Central Punjab (UCP). Currently, he is a Research Officer with the Intelligent Criminology Research Laboratory, National Center of Artificial Intelligence. His research interests include computer vision, machine learning, and deep learning.



KUSHSAIRY ABDUL KADIR (Senior Member, IEEE) received the Bachelor of Science degree in engineering from the University of West England, U.K., the Master of Science degree in mechatronics engineering from the International Islamic University of Malaysia, and the Ph.D. degree in electronic and electrical engineering from the University of Strathclyde, U.K. Prior to this, he was the Dean of the UniKL British Malaysian Institute (UniKL BMI), where he contributed significantly

to the institute's growth and development. His areas of expertise include image and signal processing, with a focus on biomedical imaging, intelligent buildings, biometrics, and rehabilitation robotics. Throughout his career, he has been actively involved in research, authoring numerous publications in esteemed journals and conference proceedings. He is currently the Deputy President of Academic and Technology with Universiti Kuala Lumpur (UniKL).

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